26900109 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECHE305 - MAIN GROUP ELEMENTS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Illustrate the diagonal relationship with an example involving an s-block element. [U] / [CO1]
2. Why does Nitrogen have a higher first ionization enthalpy than Oxygen? [U] / [CO2]
3. What are carboranes? Give one example. [K] / [CO3]
4. Give one example each of AB and AB_3 type interhalogen compounds. [K] / [CO4]
5. Explain why alkali metal cations generally exhibit a low tendency to form complex compounds. [U] / [CO1]
6. Polyphosphate is brittle at room temperature but softens on heating. Interpret using Tg. [A] / [CO2]
7. Why does boron differ significantly in properties from the rest of the Group 13 elements? [U] / [CO3]
8. Write any two uses of helium. [U] / [CO5]

9. Write the general formulae for the oxides of alkali metals and alkaline earth metals, giving one example of each. [U] / [CO1]

10. Distinguish between homoatomic and heteroatomic polymers with examples. [U] / [CO2]

11. Why do silicones show good thermal stability? [A] / [CO3]

12. What is the reason for the decrease in paramagnetic nature with an increase in the concentration of alkali metal in liquid ammonia? [U] / [CO1]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Discuss the solubility trends of sulphates of alkaline earth metals. Explain the factors responsible for the observed trends. [U] / [CO1]

14. Apply the concept of absence of d-orbitals and small size to explain why oxygen exists as a diatomic gas (O_2) while sulphur exists as polyatomic solid (S_8). [A] / [CO2]

15. Describe the reaction of diborane with ammonia (NH_3) under different temperature conditions, explaining how structural variations in the products lead to the formation of inorganic benzene (borazine). [U] / [CO3]

16. Describe the structure of ICl_3 and mention its important structural features. [U] / [CO4]

17. Discuss the properties of XeF_6. [An] / [CO5]

18. Diamond is used for cutting glass and graphite is used as a lubricant. Examine the statement based on their bonding and structure. [A] / [CO2]

19. Compare natural and synthetic zeolites with respect to their composition, structure and properties. [An] / [CO3]

20. Write the common oxidation states shown by halogens with suitable examples. [U] / [CO4]

[5x6 = 30]

Part C

Short Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Give a detailed account of the reducing nature of s-block elements. Explain the factors affecting reducing power and discuss its variation down Groups 1 and 2 with suitable examples. [U] / [CO1]

22. Define borazine. Give its preparation, uses and reason why it is called inorganic benzene. [K] / [CO2]

23. Discuss the linear combination of atomic orbitals (LCAO) approach to explain the qualitative Molecular Orbital diagram for the B−H−B bridge bond in diborane. [U] / [CO3]

24. Discuss the shapes of XeF_2, XeF_4, XeF_6 and XeO_3 using VSEPR theory. [A] / [CO5]

[10x2 = 20]